

Likelihood-Ranked Synthetic Selection in Self-Consuming Diffusion: Minority Extinction Ordering on Imbalanced Mixtures

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Abstract

Self-consuming training loops—retraining generative models on their own outputs—can induce *model collapse*, including loss of rare modes. We study unlabeled selection of synthetic samples by a denoising likelihood proxy in self-consuming DDPM training on imbalanced mixtures. Define the minority-mass ratio $r = m^{(1)}/m^{(0)}$. On a primary matrix of eight seed–imbalance cells, we report results **stratified by imbalance ρ** (not a single pooled test as the headline claim). At $\rho=5$ ($n=5$ seeds), generation 1 minority mass obeys bottom $>$ rand $>$ top on all seeds; paired retention contrasts vs. random- k have large effects (top–rand: Cohen’s $d_z=-1.96$, bootstrap 95% CI $[-0.37, -0.15]$; bottom–rand: $d_z=+1.65$, CI $[0.16, 0.43]$). Exact Wilcoxon signed-rank $p=0.0625$ for both (the minimum two-sided exact p achievable at $n=5$ when all signs agree). At $\rho=10$ ($n=3$), the full order holds on 0/3 seeds and contrasts are near null. A hypothesized “false stability” pattern—top- k improving sliced W_2 —is not observed at generation 1 ($\Delta W_2 > 0$ on 8/8 cells; exact Wilcoxon $p=0.0078$; Holm-adjusted across five generations $p=0.039$). Directional transfers (8D GMM, Digits-PCA; 3 seeds) match the $\rho=5$ order. Code: <https://github.com/muhammadalahamzeb/likelihood-rank-self-consuming-diffusion>.

1 Introduction

As synthetic data proliferates, generative models are increasingly trained on mixtures of real and model-generated samples. Iterated self-consumption can cause *model collapse* [Shumailov et al., 2023, Alemohammad et al., 2023, Gerstgrasser et al., 2024]. Mitigations include accumulating real data [Gerstgrasser et al., 2024] and filtering synthetics [Feng et al., 2024, Cai et al., 2025]. When the only score is the generator’s own likelihood, how should one select synthetics? We isolate an *interventional* selection policy at matched synthetic count and matched real-mix α , with mode-exact metrics.

What is new. A controlled top- k / random- k / bottom- k contrast under matched budgets—not a new named training algorithm or ImageNet-scale SOTA method.

Contributions. (1) Matched-budget DDPM protocol with exact mode masses. (2) Likelihood-rank ordering of minority retention, **stratified by ρ** . (3) Evidence that top- k does not improve sliced W_2 vs. random- k . (4) Directional 8D GMM and Digits-PCA checks.

2 Related work

Shumailov et al. [2023], Alemohammad et al. [2023] document collapse and quality-biased sampling (*MAD*). Gerstgrasser et al. [2024] show accumulating real+synthetic data can avoid collapse. Feng

Table 1: Primary cells. Seeds 3–4 only at $\rho=5$.

$\rho \backslash$ seed	0	1	2	3	4
5	✓	✓	✓	✓	✓
10	✓	✓	✓	—	—

et al. [2024] study task verification of labeled synth; Cai et al. [2025] (*LSF*) filter via latent probes. Inference-time temperature methods [Xu et al., 2025, Li et al., 2026] reshape sampling diversity; we intervene on training-set selection.

3 Method

3.1 Self-consuming loop

Train DDPM $f^{(g)}$ on $\mathcal{D}^{(g)}$, sample synthetics, mix fixed real buffer (fraction α) with selected synthetics.

3.2 Likelihood proxy and policies

Score $s(x)$: mean denoising MSE over a t -grid (lower = higher likelihood).

- **top- k / bottom- k / random- k** : pool of size $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}} N \rfloor)$ by lowest / highest / uniform scores; then draw n_{syn} and mix with α real.
- **mix**: same α , but synthetic portion is a uniform subsample of all N synthetics (no rank filter).
- **replace**: pure synthetic.

Thus mix $\neq$ random- k .

3.3 Metrics

Nearest-mean mode masses; minority-mean m ; ratio $r = m^{(1)}/m^{(0)}$ (“retention,” allowed > 1). Sliced W_2 : mean squared 1D Wasserstein over **24** projections; ≤ 400 synth and ≤ 400 held-out real points. Hypothesis tests use the **exact** Wilcoxon signed-rank distribution over sign assignments (enumeration; midranks for magnitude ties), not only a normal approximation.

4 Experiments

4.1 Primary design ($n=8$ cells)

$K=4$, $G=5$, $\alpha=0.5$, $k_{\text{frac}}=0.5$. Logs: `w2_fs.jsonl`.

4.2 Retention: stratified primary analysis

Because $\rho=5$ and $\rho=10$ behave differently, we do **not** treat a single pooled Wilcoxon as the primary claim. Table 2 gives means and seed-wise order counts; Table 3 gives stratified paired contrasts.

At $\rho=5$, all five seeds agree in sign; exact two-sided $p=0.0625$ is the smallest attainable at $n=5$ under the signed-rank null when all signs agree (so lack of $p<0.05$ is a discrete- n limit, not a null

Table 2: Per- ρ mean r and seed-wise counts (generation 1).

ρ	n	bottom	rand	top	Full order	top<rand / bot>rand
5	5	0.737	0.456	0.183	5/5	5/5 / 5/5
10	3	0.296	0.199	0.154	0/3	2/3 / 1/3

Table 3: Stratified paired retention contrasts vs. random- k (exact Wilcoxon; bootstrap 95% CI of mean Δ).

ρ	Contrast	Mean Δ	d_z	95% CI	Exact p
5	top–rand	−0.273	−1.96	[−0.37, −0.15]	0.0625
5	bottom–rand	+0.281	+1.65	[0.16, 0.43]	0.0625
10	top–rand	−0.045	−0.28	[−0.19, 0.13]	0.75
10	bottom–rand	+0.097	+0.38	[−0.08, 0.39]	1.00

effect— $|d_z|>1.6$). At $\rho=10$, minorities are near floor at $g=0$ and contrasts are consistent with null. For completeness, a pooled $n=8$ exact Wilcoxon yields $p=0.039$ for both contrasts, but we treat that as secondary because it mixes heterogeneous strata.

4.3 Sliced W_2

Generation-wise tests are repeated measures on the same cells (serially correlated), so Holm is a conservative acknowledgment rather than an assumption of independence. Evidence against top- k improving W_2 is strongest at $g=1$.

4.4 Real-mix α ablation (illustrative; $n=2$)

Preliminary only: at $\alpha=0.25$, means rank top worst but seed-wise order holds 1/2; at $\alpha=0.75$, top vs. random reverses on average.

4.5 Transfer checks (directional; $n=3$)

8D GMM: 0.55 ± 0.20 / 0.27 ± 0.06 / 0.08 ± 0.03 ; order 3/3. Digits-PCA: 1.49 ± 0.11 / 1.11 ± 0.08 / 0.68 ± 0.17 ; order 3/3.

4.6 Selection-pool size k_{frac} (illustrative; $n=2$)

Top- k remains worst on means at all three settings; at 0.75, bottom vs. random flips and the full order holds on 0/2 seeds.

5 Analysis

Top- k retains high-likelihood (typically majority) synthetics, reducing minority inflow. The ordering is clear when minorities remain measurable ($\rho=5$) and collapses toward noise when they are near floor ($\rho=10$). Sliced W_2 is a poor monitor of minority survival here.

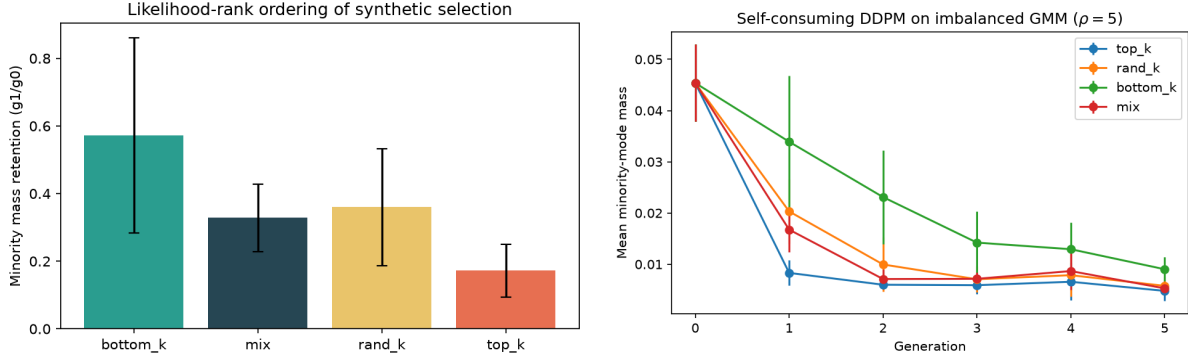


Figure 1: Left: pooled retention bars. Right: minority mass vs. generation at $\rho=5$.

Table 4: $\Delta W_2(\text{top} - \text{rand})$ by generation on all eight cells (positive = top worse). Exact Wilcoxon p ; Holm adjustment across the five (non-independent) generation-wise tests; bootstrap 95% CI of the mean.

g	Mean	Std	d_z	95% CI	$\#\{> 0\}$	Exact p (Holm)
1	+0.113	0.099	1.14	[0.05, 0.18]	8/8	0.0078 (0.039)
2	+0.061	0.060	1.00	[0.02, 0.10]	6/8	0.039 (0.094)
3	+0.039	0.035	1.11	[0.02, 0.06]	7/8	0.023 (0.094)
4	+0.027	0.032	0.83	[0.01, 0.05]	7/8	0.023 (0.094)
5	+0.024	0.033	0.73	[0.01, 0.05]	7/8	0.023 (0.094)

6 Limitations

Toy GMMs / Digits-PCA; not ImageNet-scale. At $\rho=5$, exact Wilcoxon cannot beat $p=0.0625$ with $n=5$ all-agreeing signs—inference leans on d_z and unanimous direction. α and k_{frac} ablations use two seeds (illustrative). Transfers: three seeds, no formal tests. Contribution is a controlled empirical contrast relative to MAD/LSF/Feng.

Note on double-blind review. The public GitHub URL reveals author identity. For double-blind venues, provide an anonymous code mirror; for arXiv, the named repository is appropriate.

7 Conclusion

Likelihood ranking induces a minority-retention order that is stable at moderate imbalance ($\rho=5$) and not seed-wise reliable at $\rho=10$ in our matrix. Top- k does not improve sliced W_2 vs. random- k at generation 1 on the eight-cell matrix. Preferring “most likely” synthetics is not a free lunch for minority survival when rare modes remain measurable.

References

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Table 5: Mean r by α (seeds $\{0, 1\}$, $\rho=5$). “Order holds” = bottom>rand>top; not guaranteed on both seeds except at $\alpha=0.5$.

α	bottom	rand	top	Order holds
0.25	0.25	0.17	0.05	1/2
0.50	0.62	0.49	0.24	2/2
0.75	1.34	0.77	0.94	0/2

Table 6: k_{frac} ablation (seeds $\{0, 1\}$, $\rho=5$). Under $k = \max(n_{\text{syn}}, \lfloor k_{\text{frac}}N \rfloor)$, 0.25 and 0.5 are identical here (n_{syn} binds).

k_{frac}	bottom	rand	top	Order holds
0.25	0.62	0.49	0.24	1/2
0.50	0.62	0.49	0.24	1/2
0.75	0.48	0.61	0.22	0/2

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A Reproducibility

`python scripts/reproduce_w2.py`; stats tables via `experiments/scripts/make_submission_stats.py`.
Code: <https://github.com/muhammadalahzeb/likelihood-rank-self-consuming-diffusion>.